Airline Tweet Analysis

Text search, sentiment and geospatial patterns in Elasticsearch

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Dataset: February 2015 · Verified: 28 October 2026 · Elasticsearch 9.5.5

14,640	14,485	62.74%
SOURCE ROWS	INDEX DOCUMENTS	NEGATIVE SENTIMENT

This study loads airline sentiment tweets into a strict Elasticsearch index and answers 14 search, aggregation and geographic questions. Negative tweets account for 9,088 of 14,485 documents. United has the largest volume, while the busiest day is 22 February. All query results below come from a successful live run; only 5.85% of documents have a valid location.

A1 Dataset and preparation

CSV inspection	Rows
Total rows	14,640
Non-empty tweet_coord	1,019
Placeholder [0.0, 0.0]	164
Valid coordinate rows	855

Explicit mapping & cleaning

One shard, zero replicas and **dynamic: strict**. Text uses the English analyzer; airline, sentiment and hashtags are keywords. Dates use ISO 8601 and coordinates use **geo_point**. HTML is unescaped, tags are lowercased, and empty cells are omitted.

The CSV contains 155 repeated-ID surplus rows. Using tweet_id as _id replaces earlier copies, leaving 14,485 documents and 847 unique geotagged tweets.

B1 Observed indexing output

```
Connected to cluster 'docker-cluster', Elasticsearch 9.5.5
Rows read: 14640, skipped: 0, indexed: 14640, failed: 0, with valid coordinates: 855
Documents now in 'tweets': 14485

GET /tweets/_count → {"count": 14485, "_shards":
{"total": 1, "successful": 1, "skipped": 0, "failed": 0}}
```

Live evidence: successful workflow run. The sample dry run also read 5 rows, skipped 0, and kept 2 valid locations.

B2 Why these choices?

The Bulk API groups 500 indexing actions per request, reducing network overhead. Using tweet_id as _id lets later rows and repeated runs update the same document. The point [0.0, 0.0] is removed because it is a missing-location placeholder that would distort distance and map results. Tweets without a valid point are kept with has_location set to false.

Search results

Exact filters, analyzed text and ranked tweet examples

Q1

9,088

negative tweets

2,633 are labelled United*

Q3

91

positive phrase matches

English-analyzed “customer service”

Q4

25

Delta late-flight complaints

23 February 2015 · Pacific time

*Q1 filters the dataset's airline label. Q4 filters negative sentiment, Delta and Late Flight within [23 Feb 00:00, 24 Feb 00:00), America/Los_Angeles. Original airline labels are retained.

Q2 Lost or delayed baggage

231 matches. Two required match clauses combine a baggage term with a loss or delay term. The five most recent hits are shown below; ties use tweet_id ascending.

Airline	Sentiment	Tweet text
American	negative	@AmericanAir Your response could have made all the difference. It could have made the situation better. NO TRUST...GET LOST like my bag.
American	negative	@AmericanAir this delayed bag was for my friend Lisa Pafe. She got her bag after 3 days in Costa Rica. Issue no updates on your system.
US Airways	negative	@USAirways my bag is still lost and I can not get anyone on the phone to assist me and the tracking updates have stopped
American	negative	@AmericanAir Not giving you a hard time...Just looking for basic customer service after AA lost my bag. ETA on it's return, please????????
Southwest	negative	@SouthwestAir do you know that your lost baggage portal http://t.co/hUcLXluV5h doesn't work on mobile? It's a 404 http://t.co/O4ZR27Qpcr

Original wording with whitespace normalized. English analysis matches terms even when they are not adjacent.

Q5 The five most-retweeted tweets

Airline	Retweets	Tweet excerpt
US Airways	44	@USAirways 5 hr flight delay and a delay when we land . Is that even real life ? Get me off this plane , I...
US Airways	32	@USAirways of course never again tho . Thanks for tweetin ur concern but not Doin anythin to fix what...
Delta	31	STOP. USING.THIS.WORD. IF. YOU'RE. A. COMPANY. RT @JetBlue: Our fleet's on fleek. http://t.co/Fd2TNYcTrB
US Airways	28	@USAirways with this livery back in the day. http://t.co/EEqWVAMmiy
Southwest	22	@SouthwestAir beautiful day in Seattle! http://t.co/iqu0PPVq2S

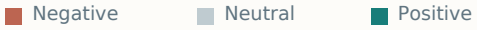
Sorted by retweet_count descending, then tweet_id ascending. The requested _source fields are text, airline and retweet_count.

Patterns across airlines

Counts describe this tweet sample, using the original sentiment labels.

Q6 Tweet volume and sentiment

Airline	Total	Negative	Neutral	Positive	Sentiment share
United	3,822	2,633	697	492	
US Airways	2,913	2,263	381	269	
American	2,604	1,870	429	305	
Southwest	2,420	1,186	664	570	
Delta	2,222	955	723	544	
Virgin America	504	181	171	152	

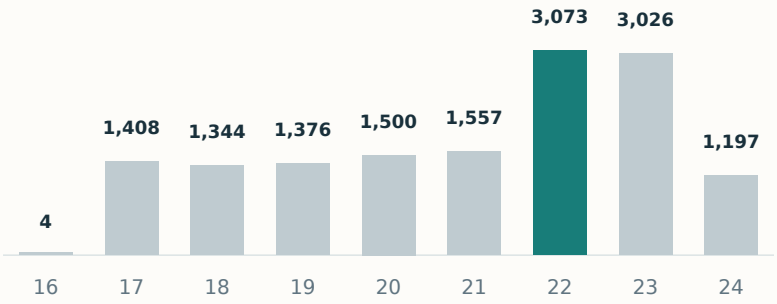


United has the most tweets (3,822). US Airways has the highest negative share (77.69%).

Q7 Top three negative reasons per airline

Airline	1st reason	2nd reason	3rd reason
United	Customer Service Issue **(681)**	Late Flight **(525)**	Can't Tell **(379)**
US Airways	Customer Service Issue **(811)**	Late Flight **(453)**	Can't Tell **(246)**
American	Customer Service Issue **(741)**	Late Flight **(237)**	Cancelled Flight **(231)**
Southwest	Customer Service Issue **(391)**	Cancelled Flight **(162)**	Can't Tell **(159)**
Delta	Late Flight **(269)**	Customer Service Issue **(199)**	Can't Tell **(186)**
Virgin America	Customer Service Issue **(60)**	Flight Booking Problems **(28)**	Can't Tell **(22)**

Q8 Daily activity



February 2015 · Pacific time

22 February was busiest: **3,073 tweets**. The first and last dates are partial days in this dataset.

Q9 Top hashtags

Hashtag	Docs
#destinationdragons	81
#fail	67
#jetblue	48
#unitedairlines	45
#customerservice	36
#usairways	28
#neveragain	27
#americanairlines	26
#united	26
#usairwaysfail	26

Where tweets originate

Location coverage is sparse; geographic findings apply to geotagged tweets only.

Q10 Location coverage

847 of 14,485 documents (5.85%) have a valid geo_point. The remaining 13,638 tweets are retained for text and sentiment analysis.

Q11 Within 25 km of JFK

83 hits around (40.6413, -73.7781). First five results, sorted by arc distance:

Distance (km)	Airline	Sentiment	Tweet ID
0.293	Delta	negative	568611474416906240
0.423	Virgin America	negative	569967019958730753
0.471	Delta	negative	568216703055876096
0.471	Delta	negative	568204705861652480
0.472	Delta	negative	568196466503770112

Q12 California box

80 negative tweets. United leads with 43.

Airline	Negative
United	43
American	13
Southwest	13
US Airways	6
Delta	4
Virgin America	1

Box: 32.5° to 42.0° N; 124.5° to 114.1° W.

Q13 O'Hare distance rings

Origin: (41.9742, -87.9073). All 847 geotweets.

Ring	Tweets	Share
0 ≤ km < 50	47	5.55%
50 ≤ km < 500	24	2.83%
km ≥ 500	776	91.62%

776 tweets (91.62%) are at least 500 km away. Range upper bounds are exclusive; the last bucket includes 500 km.

Q14 Busiest map cells and negative centroid

Geotile precision 6 (zoom/x/y), ranked by document count:

6/18/24: 205 · **6/17/25**: 53 · **6/16/23**: 53 · **6/15/26**: 46 · **6/11/25**: 46

Negative centroid: **(37.163571, -87.381890)** · 567 geotagged negative tweets

Sources & reproducibility. [1] CrowdFlower, Twitter US Airline Sentiment (2015), CC BY-NC-SA 4.0. [2] Elastic reference: match query; geo-centroid. [3] Submitted queries and saved live responses. Verified 28 Oct 2026; source commit 9d19eaf. Centroid is the coordinate mean, not an airport or a complaint-rate measure.